

2

Formulation of Two- and Three-dimensional Boundary-value Problems

2.1 INTRODUCTORY REMARKS

It makes no sense to attempt to “solve” a boundary-value problem without a precise knowledge of what the problem is. The truth of this statement seems self-evident. Unfortunately, attempts are often made to solve vaguely defined problems, creating considerable confusion and, sometimes, totally erroneous results. In this chapter we present precise statements of multidimensional boundary-value problems in classical linear heat conduction and elastostatics. The presentation is similar in many respects to that for the one-dimensional model problem of Chapter 1. In particular, we discuss strong and weak forms, their equivalence, corresponding Galerkin formulations, the definitions of element arrays, and pertinent data processing concepts. In multi-dimensions, the data processing ideas necessarily become more involved. The reader is urged to study them carefully as they are necessary in order to understand the computer implementation of finite element techniques.

2.2 PRELIMINARIES

Let n_{sd} ($= 2$ or 3) denote the *number of space dimensions* of the problem under consideration. Let $\Omega \subset \mathbb{R}^{n_{sd}}$ be an open set¹ with piecewise smooth boundary Γ . A general point in $\mathbb{R}^{n_{sd}}$ is denoted by x . We will identify the point x with its position vector emanating from the origin of $\mathbb{R}^{n_{sd}}$. The *unit outward normal vector to Γ* is denoted by n .

¹ For our purposes, it is sufficient to think of an open set as one without its boundary.

We shall employ the following alternative representations for $\mathbf{x}$ and $\mathbf{n}$:

$$(n_{sd} = 2): \quad \mathbf{x} = \{x_i\} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} x \\ y \end{Bmatrix} \quad \mathbf{n} = \{n_i\} = \begin{Bmatrix} n_1 \\ n_2 \end{Bmatrix} = \begin{Bmatrix} n_x \\ n_y \end{Bmatrix} \quad (2.2.1)$$

$$(n_{sd} = 3): \quad \mathbf{x} = \{x_i\} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} x \\ y \\ z \end{Bmatrix} \quad \mathbf{n} = \{n_i\} = \begin{Bmatrix} n_1 \\ n_2 \\ n_3 \end{Bmatrix} = \begin{Bmatrix} n_x \\ n_y \\ n_z \end{Bmatrix} \quad (2.2.2)$$

where x_i and n_i , $1 \leq i \leq n_{sd}$, are the Cartesian components of $\mathbf{x}$ and $\mathbf{n}$, respectively; see Figure 2.2.1. Unless otherwise specified we shall work in terms of Cartesian components of vectors and tensors.

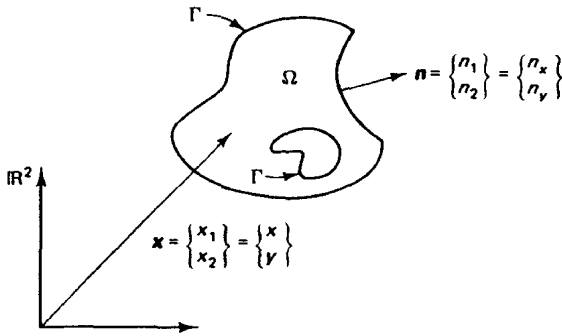


Figure 2.2.1

We assume that Γ admits the decomposition

$$\Gamma = \overline{\Gamma_q \cup \Gamma_k} \quad (2.2.3)$$

where

$$\Gamma_q \cap \Gamma_k = \emptyset \quad (2.2.4)$$

and Γ_q and Γ_k are open sets in Γ . The notations are defined as follows: $\cup$ is the **set union** symbol. Thus $\Gamma_q \cup \Gamma_k$ means the set of all points $\mathbf{x}$ contained in either Γ_q or Γ_k . Also, $\cap$ is the **set intersection** symbol. Thus $\Gamma_q \cap \Gamma_k$ means the set of all points contained in both Γ_q and Γ_k . The **empty set** is denoted by $\emptyset$. Thus (2.2.4) means that there is no point $\mathbf{x}$ contained in both Γ_q and Γ_k (i.e., Γ_q and Γ_k do not intersect or overlap). A bar above a set means **set closure**, i.e., the union of the set with its boundary. Thus

$$\overline{\Omega} = \Omega \cup \Gamma \quad (2.2.5)$$

To understand the meaning of $\overline{\Gamma_q \cup \Gamma_k}$, we must define the boundaries of Γ_q and Γ_k in Γ . We shall do this with the aid of an example.

Example 1

Let $\Omega = \{\mathbf{x} \in \mathbb{R}^2 \mid x^2 + y^2 < 1, \text{ i.e., the interior of the unit disc}\}$. The boundary of Ω is $\Gamma = \{\mathbf{x} \in \mathbb{R}^2 \mid x^2 + y^2 = 1, \text{ i.e., the unit circle}\}$. Let

$$\Gamma_q = \Gamma \cap \{\mathbf{x} \in \mathbb{R}^2 \mid y > 0\} \quad (2.2.6)$$

The “boundary of Γ_ϱ ” consists of the endpoints of the upper semicircle, i.e., $\{x \in \mathbb{R}^2 \mid x = -1, y = 0, \text{ and } x = +1, y = 0\}$. Thus

$$\overline{\Gamma_\varrho} = \Gamma \cap \{x \in \mathbb{R}^2 \mid y \geq 0\} \quad (2.2.7)$$

Similarly, let

$$\Gamma_k = \Gamma \cap \{x \in \mathbb{R}^2 \mid y < 0\} \quad (2.2.8)$$

Thus

$$\overline{\Gamma_k} = \Gamma \cap \{x \in \mathbb{R}^2 \mid y \leq 0\} \quad (2.2.9)$$

Clearly

$$\overline{\Gamma_\varrho \cup \Gamma_k} = \overline{\Gamma_\varrho} \cup \overline{\Gamma_k} = \overline{\Gamma} \quad (2.2.10)$$

These sets are depicted in Fig. 2.2.2.

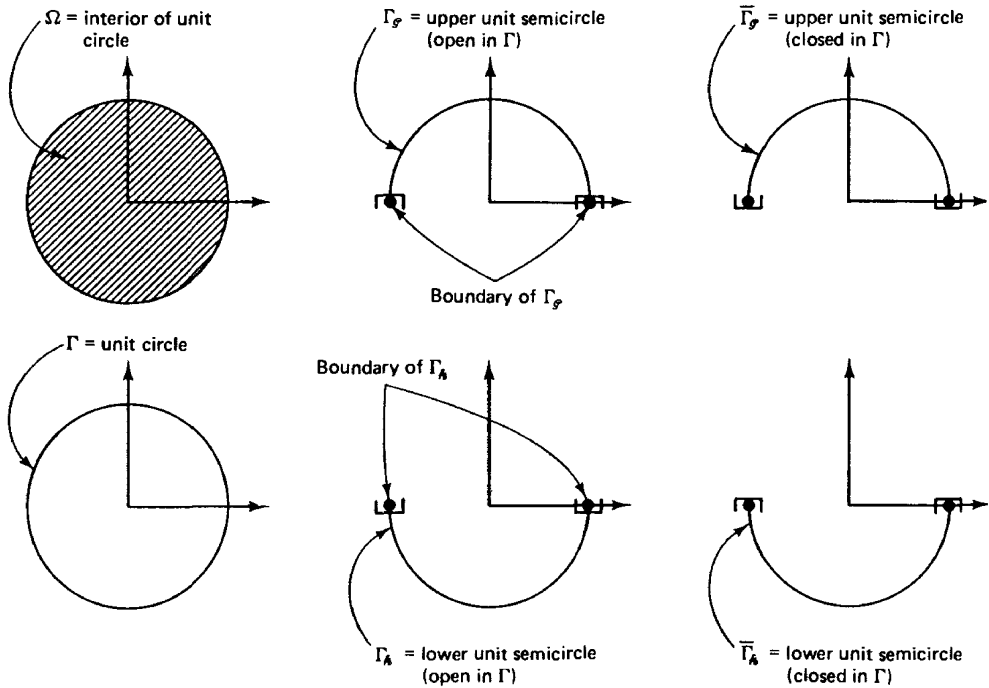


Figure 2.2.2

We shall assume throughout that $\Gamma_\varrho \neq \emptyset$ but allow for the case $\Gamma_k = \emptyset$.

Let the indices i, j, k, l , run over the values $1, \dots, n_{sd}$. Differentiation is denoted by a comma (e.g., $u_{,i} = u_{,x_i} = \partial u / \partial x_i$) and repeated indices imply summation (e.g., in $\mathbb{R}^3$, $u_{,ii} = u_{,11} + u_{,22} + u_{,33} = \partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 + \partial^2 u / \partial z^2$). The summation convention only applies to the indices i, j, k , and l and only to two repeated indices. If there are three, or more, repeated indices in an expression, then

the summation convention is *not* in effect. (This is in keeping with the usual convention.)

Divergence theorem. Let $f: \bar{\Omega} \rightarrow \mathbb{R}$ be C^1 . Then

$$\int_{\Omega} f_{,i} d\Omega = \int_{\Gamma} f n_i d\Gamma \quad (2.2.11)$$

The proof may be found in [1].

Integration by parts. Let f be as above and also let $g: \bar{\Omega} \rightarrow \mathbb{R}$ be C^1 . Then

$$\int_{\Omega} f_{,i} g d\Omega = - \int_{\Omega} f g_{,i} d\Omega + \int_{\Gamma} f g n_i d\Gamma \quad (2.2.12)$$

Proof. We integrate the identity (i.e., “product rules of differentiation,” see [1])

$$(fg)_{,i} = f_{,i} g + f g_{,i}$$

to get

$$\int_{\Omega} (fg)_{,i} d\Omega = \int_{\Omega} f_{,i} g d\Omega + \int_{\Omega} f g_{,i} d\Omega$$

and then use the divergence theorem to convert the left-hand side into a boundary integral. ■

2.3 CLASSICAL LINEAR HEAT CONDUCTION: STRONG AND WEAK FORMS; EQUIVALENCE

Let q_i denote (Cartesian components of) the *heat flux vector*, let u be the *temperature*, and let ℓ be the *heat supply per unit volume*. Assume the heat flux vector is defined in terms of the temperature gradient by the *generalized Fourier law*²:

$$q_i = -\kappa_{ij} u_{,j}, \quad \kappa_{ij} = \kappa_{ji} \quad (\text{symmetry}) \quad (2.3.1)$$

where the *conductivities*, κ_{ij} 's, are given functions of $\mathbf{x}$. (If the κ_{ij} 's are constant throughout Ω , the body is said to be *homogeneous*.) The *conductivity matrix*, $\boldsymbol{\kappa} = [\kappa_{ij}]$, is assumed positive definite (see the definition in Sec. 1.9). The most common situation in practice is the *isotropic case* in which $\kappa_{ij}(\mathbf{x}) = \kappa(\mathbf{x})\delta_{ij}$, where δ_{ij} is the Kronecker delta.

²The generalized Fourier law is a *constitutive equation*, or *equation of state*, which reflects the heat conduction properties of the body (i.e., Ω) under consideration.

A formal statement³ of the strong form of the boundary-value problem is as follows:

$$(S) \left\{ \begin{array}{l} \text{Given } \ell : \Omega \rightarrow \mathbb{R}, q : \Gamma_q \rightarrow \mathbb{R} \text{ and } h : \Gamma_h \rightarrow \mathbb{R}, \text{ find } u : \bar{\Omega} \rightarrow \mathbb{R} \text{ such that} \\ \qquad q_{i,i} = \ell \quad \text{in } \Omega \quad (\text{heat equation}) \quad (2.3.2) \\ \qquad u = q \quad \text{on } \Gamma_q \quad (2.3.3)^4 \\ \qquad -q_i n_i = h \quad \text{on } \Gamma_h \quad (2.3.4) \\ \text{where } q_i \text{ is defined by (2.3.1).} \end{array} \right.$$

The functions q and h are the *prescribed boundary temperature and heat flux*, respectively. This problem possesses a unique solution for appropriate restrictions on the given data.

In more mathematical terminology, (2.3.2) is a generalized Poisson equation, (2.3.3) is a Dirichlet boundary condition, and (2.3.4) is a Neumann boundary condition.

We shall now construct a weak formulation of the boundary-value problem analogous to that for the one-dimensional problem of Chapter 1. In particular, (2.3.3) and (2.3.4) will be treated as essential and natural boundary conditions, respectively. As before, let $\mathcal{S}$ denote the trial solution space and $\mathcal{U}$ the variation space. This time $\mathcal{S}$ and $\mathcal{U}$ consist of real-valued functions defined on $\bar{\Omega}$ satisfying certain smoothness requirements, such that all members of $\mathcal{S}$ satisfy (2.3.3), whereas if $w \in \mathcal{U}$, then

$$w = 0 \quad \text{on } \Gamma_q \quad (2.3.5)$$

The weak formulation of the problem goes as follows:

$$(W) \left\{ \begin{array}{l} \text{Given } \ell : \Omega \rightarrow \mathbb{R}, q : \Gamma_q \rightarrow \mathbb{R} \text{ and } h : \Gamma_h \rightarrow \mathbb{R}, \text{ find } u \in \mathcal{S} \text{ such that} \\ \text{for all } w \in \mathcal{U} \\ \qquad \boxed{-\int_{\Omega} w_{,i} q_i d\Omega = \int_{\Omega} w \ell d\Omega + \int_{\Gamma_h} w h d\Gamma} \quad (2.3.6) \\ \text{where } q_i \text{ is defined by (2.3.1).} \end{array} \right.$$

Theorem. Assume all functions involved are smooth enough to justify the manipulations. Then a solution of (S) is a solution of (W) and vice versa.

Proof. 1. Assume u is the solution of (S). By virtue of (2.3.3), $u \in \mathcal{S}$. Pick any $w \in \mathcal{U}$ and proceed as follows:

³ By a formal statement, we mean one in which we do not precisely delineate the spaces to which the functions involved belong.

⁴ The statement $u = q$ on Γ_q means $u(x) = q(x)$ for all $x \in \Gamma_q$, and so on.

$$0 = \int_{\Omega} w(\underbrace{q_{i,i} - \ell}_0) d\Omega \quad (\text{heat equation, i.e., (2.3.2)})$$

$$= -\int_{\Omega} w_{,i} q_i d\Omega + \int_{\Gamma} w q_i n_i d\Gamma - \int_{\Omega} w \ell d\Omega \quad (\text{integration by parts})$$

$$= -\int_{\Omega} w_{,i} q_i d\Omega - \int_{\Gamma_k} w h d\Gamma - \int_{\Omega} w \ell d\Omega \quad (\text{generalized Fourier law, } w = 0 \text{ on } \Gamma_q, \text{ and heat flux boundary condition, i.e. (2.3.4)})$$

Therefore (2.3.6) is satisfied and so u is a solution of (W) .

2. Assume u is the solution of (W) . Then $u = q$ on Γ_q , and for all $w \in \mathcal{U}$

$$\begin{aligned} 0 &= \int_{\Omega} w_{,i} q_i d\Omega + \int_{\Omega} w \ell d\Omega + \int_{\Gamma_k} w h d\Gamma \quad (\text{by (2.3.6)}) \\ &= \int_{\Omega} w(-q_{i,i} + \ell) d\Omega + \int_{\Gamma_k} w(n_i q_i + h) d\Gamma \end{aligned} \quad (2.3.7)$$

Let

$$\alpha = -q_{i,i} + \ell$$

$$\beta = q_i n_i + h$$

To show (2.3.2) and (2.3.4) are satisfied and thus complete the proof, we must prove that

$$\alpha = 0 \quad \text{on } \Omega$$

$$\beta = 0 \quad \text{on } \Gamma_k$$

First pick $w = \alpha\phi$ where

- i. $\phi > 0$ on Ω ;
- ii. $\phi = 0$ on Γ ; and
- iii. ϕ is smooth.

(These conditions insure that $w \in \mathcal{U}$.) With this choice for w , (2.3.7) becomes

$$0 = \int_{\Omega} \alpha^2 \phi d\Omega$$

which implies $\alpha = 0$ on Ω .

Now pick $w = \beta\psi$, where

- i'. $\psi > 0$ on Γ_k ;
- ii'. $\psi = 0$ on Γ_q ; and
- iii'. ψ is smooth.

(These conditions insure that $w \in \mathcal{U}$.) With this choice for w , (2.3.7) becomes

(making use of $\alpha = 0$):

$$0 = \int_{\Gamma_k} \beta^2 \psi d\Gamma$$

from which it follows that $\beta = 0$ on Γ_k . Thus u is a solution of (S). ■

It is convenient to introduce an abstract version of (2.3.6). Let

$$a(w, u) = \int_{\Omega} w_{,i} \kappa_{ij} u_{,j} d\Omega \quad (2.3.8)$$

$$(w, \ell) = \int_{\Omega} w \ell d\Omega \quad (2.3.9)$$

$$(w, h)_{\Gamma} = \int_{\Gamma_k} w h d\Gamma \quad (2.3.10)$$

Then (2.3.6) may be written as

$$a(w, u) = (w, \ell) + (w, h)_{\Gamma} \quad (2.3.11)$$

Exercise 1. Verify that $a(\cdot, \cdot)$, $(\cdot, \cdot)$ and $(\cdot, \cdot)_{\Gamma}$, as just defined, are symmetric bilinear forms. (Note that the symmetry of $a(\cdot, \cdot)$ follows from the symmetry of the conductivities.)

In manipulating terms in theories involving vector and tensor quantities, the indicial notation used is very explicit and convenient. However, when we come to the Galerkin formulation analogous to (2.3.6), additional indices necessarily appear. The situation becomes very complicated due to the greater number of indices involved and due to the ranges of the various indices being different. When we come to elasticity theory, the situation is even worse as the corresponding terms have an even greater number of indices. For these reasons it is useful at this point to adopt an *index-free* notation for (2.3.6). Aside from stemming the proliferation of indices, we shall find later on that this formulation is conducive to the computer implementation of the element arrays, especially in more complicated situations such as elasticity.

In introducing our index-free notation we shall assume for definiteness that $n_{sd} = 2$. Let ∇ denote the gradient operator; thus

$$\nabla u = \{u_{,i}\} = \begin{Bmatrix} u_{,1} \\ u_{,2} \end{Bmatrix} \quad (2.3.12)$$

$$\nabla w = \{w_{,i}\} = \begin{Bmatrix} w_{,1} \\ w_{,2} \end{Bmatrix} \quad (2.3.13)$$

In the case of two space dimensions, the conductivity matrix may be written as

$$\kappa = [\kappa_{ij}] = \begin{bmatrix} \kappa_{11} & \kappa_{12} \\ \kappa_{21} & \kappa_{22} \end{bmatrix} \quad (\text{symmetric}) \quad (2.3.14)$$

In the isotropic case, (2.3.14) simplifies to

$$\kappa = \kappa[\delta_{ij}] = \kappa \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (2.3.15)$$

In terms of the above expressions, the integrand of (2.3.8) may be written in index-free fashion:

$$w_{,i} \kappa_{ij} u_{,j} = (\nabla w)^T \kappa (\nabla u) \quad (2.3.16)$$

Thus in place of (2.3.8) we may write

$$a(w, u) = \int_{\Omega} (\nabla w)^T \kappa (\nabla u) d\Omega \quad (2.3.17)$$

Exercise 2. Verify (2.3.16) for the cases $n_{sd} = 2$ and 3.

2.4 HEAT CONDUCTION: GALERKIN FORMULATION; SYMMETRY AND POSITIVE-DEFINITENESS OF K

Let $\mathcal{S}^h$ and $\mathcal{U}^h$ be finite-dimensional approximations to $\mathcal{S}$ and $\mathcal{U}$, respectively. We assume all members of $\mathcal{U}^h$ vanish, or vanish approximately, on Γ_q and that each member of $\mathcal{S}^h$ admits the representation

$$u^h = v^h + q^h \quad (2.4.1)$$

where $v^h \in \mathcal{U}^h$ and q^h results in satisfaction, or at least approximate satisfaction, of the boundary condition $u = q$ on Γ_q .

The Galerkin formulation is given as follows:

$$(G) \left\{ \begin{array}{l} \text{Given } f, q, \text{ and } h \text{ [as in (W)], find } u^h = v^h + q^h \in \mathcal{S}^h \text{ such that for all } \\ w^h \in \mathcal{U}^h \text{ (cf. Sec. 1.5):} \end{array} \right. \quad \boxed{a(w^h, v^h) = (w^h, f) + (w^h, h)_{\Gamma} - a(w^h, q^h)} \quad (2.4.2)$$

We now view our domain as “discretized” into element domains Ω^e , $1 \leq e \leq n_{el}$. In two dimensions the element domains might be simply triangles and quadrilaterals; see Fig. 2.4.1. Nodal points may exist anywhere on the domain but most frequently appear at the element vertices and interelement boundaries and less often in the interiors.

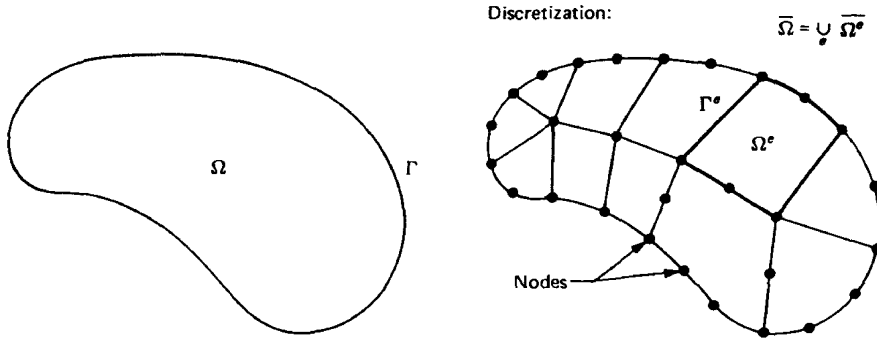


Figure 2.4.1

In Sec. 1.9 the global nodal ordering and ordering of equations in the matrix system coincided. In multidimensional applications this would prove to be an inconvenient restriction with regard to data preparation.

In what follows, a more flexible scheme is described. Let $\eta = \{1, 2, \dots, n_{np}\}$, the *set of global node numbers* where n_{np} is the *number of nodal points*. By the terminology *q-node* we shall mean a node, A , at which it is prescribed that $u^h = q$. Let $\eta_q \subset \eta$ be the set of “*q-nodes*.” The *complement of η_q in η* , denoted by $\eta - \eta_q$, is the set of nodes at which u^h is to be determined. The number of nodes in $\eta - \eta_q$ equals n_{eq} , the *number of equations*.

A typical member of $\mathcal{U}^h$ is assumed to have the form

$$w^h(x) = \sum_{A \in \eta - \eta_q} N_A(x) c_A \quad (2.4.3)$$

where N_A is the shape function associated with node number A and c_A is a constant. We assume throughout that $w^h = 0$ if and only if $c_A = 0$ for each $A \in \eta - \eta_q$. Likewise

$$v^h(x) = \sum_{A \in \eta - \eta_q} N_A(x) d_A \quad (2.4.4)$$

where d_A is the unknown at node A (i.e., temperature) and

$$q^h(x) = \sum_{A \in \eta_q} N_A(x) q_A, \quad q_A = q(x_A) \quad (2.4.5)$$

From (2.4.5), we see that q^h has been *defined* to be the nodal interpolate of q by way of the shape functions.⁵ Consequently, q^h will be, generally, only an approximation of q . See Fig. 2.4.2. Additional sources of error are (1) the use of approximations ℓ^h and h^h in place of ℓ and h , respectively; and (2) domain approximations in which the element boundaries do not exactly coincide with Γ . Analyses of these approximations are presented in Strang and Fix [2].

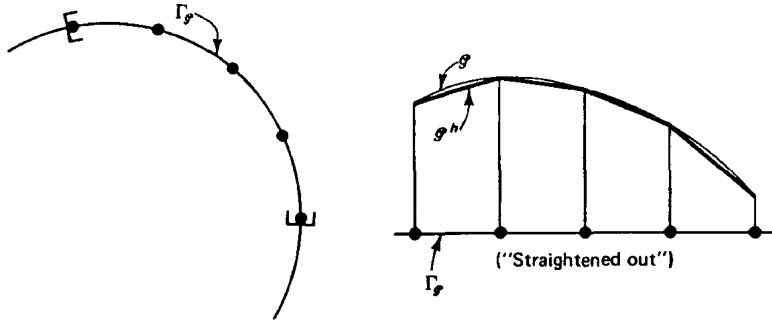


Figure 2.4.2 Piecewise linear approximation of boundary data (schematic).

Substituting (2.4.3)–(2.4.5) into (2.4.2) and arguing as in Sec. 1.6, results in

$$\sum_{B \in \eta - \eta_q} a(N_A, N_B) d_B = (N_A, \ell) + (N_A, h)_\Gamma - \sum_{\substack{B \in \eta_q \\ A \in \eta - \eta_q}} a(N_A, N_B) q_B, \quad (2.4.6)$$

To define the global stiffness matrix and force vector, we need to first specify the global ordering of equations. For this purpose we introduce the ID array, sometimes called the *destination array*, which assigns to node A the corresponding global equation number, viz.,

$$\text{ID}(A) = \begin{cases} P & \text{if } A \in \eta - \eta_q \\ 0 & \text{if } A \in \eta_q \end{cases} \quad \begin{array}{l} \text{Global equation} \\ \text{number} \end{array} \quad (2.4.7)$$

⁵This is not the only possibility, nor the best from the standpoint of accuracy. However, in practice it is generally the most convenient.

where $1 \leq P \leq n_{eq}$. The dimension of ID is n_{np} . As may be seen from (2.4.7), nodes at which q is prescribed are assigned "equation number" zero. An example of the setup of ID and other important data-processing arrays is presented in Sec. 2.6.

The matrix equivalent of (2.4.6) is given as follows:

$$Kd = F \quad (2.4.8)$$

$$K = [K_{PQ}], \quad d = \{d_Q\}, \quad F = \{F_P\}, \quad 1 \leq P, Q \leq n_{eq} \quad (2.4.9)$$

$$K_{PQ} = a(N_A, N_B), \quad P = \text{ID}(A), \quad Q = \text{ID}(B) \quad (2.4.10)$$

$$F_P = (N_A, \ell) + (N_A, h)_\Gamma - \sum_{B \in \eta_q} a(N_A, N_B) q_B \quad (2.4.11)$$

The main properties of K are established in the following theorem.

Theorem

1. K is symmetric.
2. K is positive definite.

Proof

1. The symmetry of K follows directly from the symmetry of $a(\cdot, \cdot)$, viz.,

$$\begin{aligned} K_{PQ} &= a(N_A, N_B) && \text{(by definition)} \\ &= a(N_B, N_A) && \text{(symmetry of } a(\cdot, \cdot)) \\ &= K_{QP} && \text{(by definition)} \end{aligned}$$

2. (Recall that we must show (i) $c^T K c \geq 0$ and (ii) $c^T K c = 0$ implies $c = 0$.)

To each n_{eq} -vector $c = \{c_P\}$, we may associate a member $w^h \in \mathcal{U}^h$ by the expression $w^h = \sum_{A \in \eta - \eta_q} N_A \bar{c}_A$, where $\bar{c}_A = c_P$, $P = \text{ID}(A)$.

i.

$$\begin{aligned} c^T K c &= \sum_{P, Q=1}^{n_{eq}} c_P K_{PQ} c_Q \\ &= \sum_{A, B \in \eta - \eta_q} \bar{c}_A a(N_A, N_B) \bar{c}_B \\ &= a\left(\sum_{A \in \eta - \eta_q} N_A \bar{c}_A, \sum_{B \in \eta - \eta_q} N_B \bar{c}_B\right) && \text{(bilinearity of } a(\cdot, \cdot)) \\ &= a(w^h, w^h) && \text{(definition of } w^h) \\ &= \int_{\Omega} \underbrace{w^h_i K_{ij} w^h_j}_{\geq 0} d\Omega && \text{(positive-definiteness of conductivities)} \\ &\geq 0 \end{aligned}$$

ii. Assume $\mathbf{c}^T \mathbf{K} \mathbf{c} = 0$. By the proof of part (i),

$$\int_{\Omega} \underbrace{w_{,i}^h \kappa_{ij} w_{,j}^h}_{\geq 0} d\Omega = 0$$

and thus it follows that

$$w_{,i}^h \kappa_{ij} w_{,j}^h = 0$$

By the positive-definiteness hypothesis on the conductivities, this requires $w_{,i}^h = 0$ and so w^h must be constant. However $w^h = 0$ on Γ_a (which is not empty) and so w^h must be zero throughout Ω . By the definition of w^h , it follows that each $c_p = 0$; that is $\mathbf{c} = \mathbf{0}$, which was to be proved. ■

Remarks

1. Observe that it is the positive-definiteness hypothesis on the constitutive coefficients (i.e., κ_{ij} 's) and the boundary condition incorporated in the definition of $\mathcal{U}^h$ which together result in the positive-definiteness of $\mathbf{K}$ and thus ensure its invertibility.

2. The explicit structure of the shape functions, which will be delineated in Chapter 3, will also result in $\mathbf{K}$ being banded.

Exercise 1. (This exercise is a multidimensional analog of the one contained in Sec. 1.8.) Let

$$\Gamma_{\text{int}} = \left(\bigcup_{e=1}^{n_{el}} \Gamma^e \right) - \Gamma \quad (\text{interior element boundaries})$$

One side of Γ_{int} is (arbitrarily) designated to be the “+ side” and the other is the “− side.” Let $\mathbf{n}^+$ and $\mathbf{n}^-$ be unit normals to Γ_{int} which point in the plus and minus directions, respectively. Clearly $\mathbf{n}^+ = -\mathbf{n}^-$. Let q_i^+ and q_i^- denote the values of q_i obtained by approaching $\mathbf{x} \in \Gamma_{\text{int}}$ from + and − sides, respectively. The “jump” in $q_n = q_i n_i$ at $\mathbf{x}$ is defined to be

$$\begin{aligned} [q_n] &= (q_i^+ - q_i^-) n_i^+ \\ &= q_i^+ n_i^+ + q_i^- n_i^- \end{aligned}$$

As may be easily verified, the jump is invariant with respect to reversing the + and − designations.

Consider the weak formulation (i.e., (2.3.6)) and assume w and u are smooth on the element interiors but may experience discontinuities in gradient across element boundaries. (Functions of this type contain the standard C^0 finite element interpolations; see Chapter 3.) Show that

$$0 = \sum_{e=1}^{n_{el}} \int_{\Omega^e} w(q_{i,i} - f) d\Omega - \int_{\Gamma_k} w(q_n + h) d\Gamma + \int_{\Gamma_{\text{int}}} w[q_n] d\Gamma$$

from which the Euler-Lagrange conditions may be readily deduced:

- i. $q_{i,i} = \ell$ in $\bigcup_{e=1}^{n_{el}} \Omega^e$
- ii. $-q_n = h$ on Γ_k
- iii. $[q_n] = 0$ on Γ_{int}

As may be seen, (i) is the heat equation on the *element interiors* and (iii) is a continuity condition across element boundaries on the heat flux. Contrast the present results with those obtained assuming w and u are *globally* smooth.

The Galerkin finite element formulation obtains an *approximate* solution to (i) through (iii).

2.5 HEAT CONDUCTION: ELEMENT STIFFNESS MATRIX AND FORCE VECTOR

As before, we can break up the global arrays into sums of elemental contributions:

$$\mathbf{K} = \sum_{e=1}^{n_{el}} \mathbf{K}^e, \quad \mathbf{K}^e = [\mathbf{K}_{PQ}^e] \quad (2.5.1)$$

$$\mathbf{F} = \sum_{e=1}^{n_{el}} \mathbf{F}^e, \quad \mathbf{F}^e = \{\mathbf{F}_{\hat{P}}^e\} \quad (2.5.2)$$

where

$$\mathbf{K}_{PQ}^e = a(N_A, N_B)^e = \int_{\Omega^e} (\nabla N_A)^T \kappa (\nabla N_B) d\Omega \quad (2.5.3)$$

$$\begin{aligned} \mathbf{F}_{\hat{P}}^e &= (N_A, \ell)^e + (N_A, h)_f - \sum_{B \in \eta_q} a(N_A, N_B)^e q_B \\ &= \int_{\Omega^e} N_A \ell d\Omega + \int_{\Gamma_k^e} N_A h d\Gamma - \sum_{B \in \eta_q} a(N_A, N_B)^e q_B \end{aligned} \quad (2.5.4)$$

$$\Gamma_k^e = \Gamma_k \cap \Gamma^e, \quad P = \text{ID}(A), \quad Q = \text{ID}(B) \quad (2.5.5)$$

See Fig. 2.5.1 for an illustration of Γ_k^e .

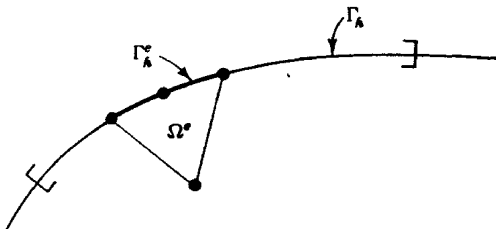


Figure 2.5.1

The element stiffness, k^e , and element force vector, f^e , may be deduced from these equations:

$$k^e = [k_{ab}^e], \quad f^e = \{f_a^e\}, \quad 1 \leq a, b \leq n_{en} \quad (2.5.6)$$

$$k_{ab}^e = a(N_a, N_b)^e = \int_{\Omega^e} (\nabla N_a)^T \kappa (\nabla N_b) d\Omega \quad (2.5.7)$$

$$f_a^e = \int_{\Omega^e} N_a q d\Omega + \int_{\Gamma_i^e} N_a h d\Gamma - \sum_{b=1}^{n_{en}} k_{ab}^e q_b^e \quad (2.5.8)$$

where (recall) n_{en} is the number of element nodes, and $q_b^e = q(x_b^e)$ if q is prescribed at node number b and equals zero otherwise.⁶

The global arrays, K and F may be formed from the element arrays k^e and f^e , respectively, by way of an assembly algorithm as described in Sec. 1.14.

The element stiffness matrix can be written in a standard form convenient for programming:

$$k^e = \int_{\Omega^e} B^T D B d\Omega \quad (2.5.9)$$

where, in the present case,

$$\underbrace{D}_{n_{sd} \times n_{sd}} = \kappa \quad (2.5.10)$$

$$\underbrace{B}_{n_{sd} \times n_{en}} = [B_1, B_2, \dots, B_{n_{en}}] \quad (2.5.11)$$

$$\underbrace{B_a}_{n_{sd} \times 1} = \nabla N_a \quad (2.5.12)$$

The component version of (2.5.9) is

$$k_{ab}^e = \int_{\Omega^e} B_a^T D B_b d\Omega \quad (2.5.13)$$

⁶ An implicit assumption in localizing the q -term is that if x_a is not a node attached to element e , then $N_a(x) = 0$ for all $x \in \Omega^e$. Otherwise, the last term in (2.5.4) may involve q -data of nodes not attached to element e , which is not accounted for in (2.5.8).

Exercise 1. Let

$$\underbrace{d^e}_{n_{en} \times 1} = \{d_a^e\} = \begin{Bmatrix} d_1^e \\ d_2^e \\ \vdots \\ d_{n_{en}}^e \end{Bmatrix} \quad (2.5.14)$$

where

$$d_a^e = u^h(x_a^e) \quad (2.5.15)^7$$

d^e is called the *element temperature vector*. Show that the heat flux vector at point $x \in \Omega^e$ can be calculated from the formula

$$q(x) = -D(x)B(x)d^e = -D(x) \sum_{a=1}^{n_{en}} B_a d_a^e \quad (2.5.16)$$

Exercise 2. Consider the strong statement of the boundary-value problem in classical linear heat conduction in which the h -type boundary condition (i.e., eq. (2.3.4)) is replaced by the following expression:

$$\lambda u - q_i n_i = h \quad \text{on } \Gamma_h \quad (2.5.17)$$

where $\lambda \geq 0$ is a given function of $x \in \Gamma_h$. Generalize the weak formulation to include (2.5.17) as a natural boundary condition. Obtain an expression for the additional contribution to k_{ab}^e (cf. (2.5.13)) arising from (2.5.17). Show that K is positive-definite.

The boundary condition (2.5.17) is equivalent to what is often called *Newton's law of heat transfer*; λ is called the *coefficient of heat transfer*. This boundary condition applies to the case in which the heat flux is proportional to the difference of the surface temperatures of the body and surrounding medium, the latter formally represented by h/λ in (2.5.17).

2.6 HEAT CONDUCTION: DATA PROCESSING ARRAYS; ID, IEN, AND LM

The element nodal data is stored in the array IEN, the *element nodes array*, which relates local node numbers to global node numbers, viz.,

$$\boxed{\begin{array}{ccc} \text{IEN}(a, e) & = & A \\ \uparrow & \uparrow & \uparrow \\ \text{Local} & \text{Element} & \text{Global} \\ \text{node} & \text{number} & \text{node} \\ \text{number} & & \text{number} \end{array}} \quad (2.6.1)$$

The relationship between global node numbers and global equation numbers as well as nodal boundary condition information is stored in the ID array (see 2.4.7). In

⁷ The q -boundary conditions are accounted for in this definition.

practice, the IEN and ID arrays are set up from input data. The LM array, which was described in the context of the one-dimensional model problem in Sec. 1.14, may then be constructed from the relation

$$\boxed{\text{LM}(a, e) = \text{ID}(\text{IEN}(a, e))} \quad (2.6.2)$$

Because of the previous relationship, we often think of LM as the element “localization” of ID. Strictly speaking the LM array is redundant. However, it is generally convenient in computing to set up LM once and for all, rather than make use of (2.6.2) repeatedly.

Example 1 illustrates the structure of the ID, IEN, and LM arrays.

Example 1

Consider the mesh of four-node, rectangular elements shown in Fig. 2.6.1. We assume that the local node numbering begins at the lower left-hand node of each element and proceeds in counterclockwise fashion. This is illustrated in Fig. 2.6.1 for element 2, which is typical. We also assume that essential boundary conditions (i.e., “ q -type”) are specified at nodes 1, 4, 7, and 10. Thus there will only be eight equations in the global system $Kd = F$. We adopt the usual convention that the global equation numbers run in ascending order with respect to the ascending order of global node numbers. The ID, IEN, and LM arrays are given in Fig. 2.6.2. The reader is urged to verify the details.

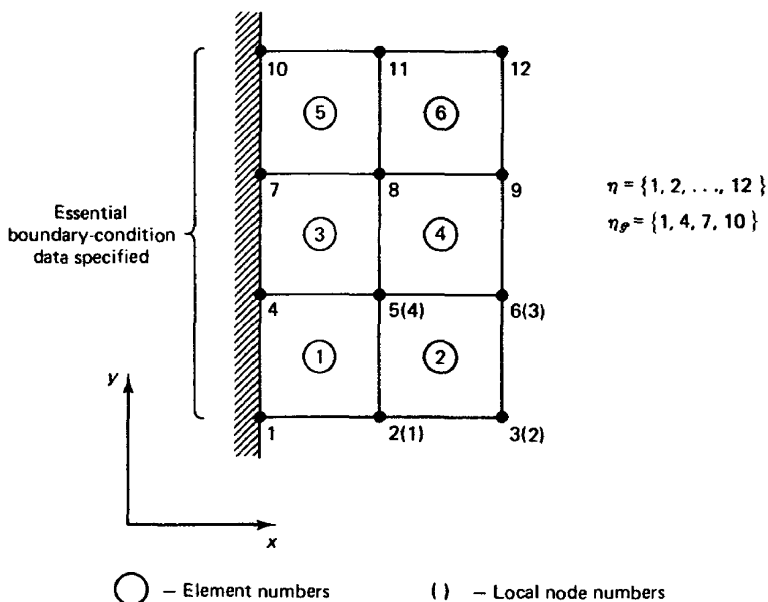


Figure 2.6.1 Mesh of four-node, rectangular elements; global and local node numbers, element numbers, and essential boundary condition nodes.

ID array:

Global node numbers (A)												$(n_{np} = 12)$
1	2	3	4	5	6	7	8	9	10	11	12	
0*	1	2	0	3	4	0	5	6	0	7	8	

$(n_{eq} = 8)$

$$P = ID(A)$$

IEN array:

		Element numbers (e)						
		123456						$(n_{el} = 6)$
Local node numbers (a)	1	1	2	4	5	7	8	
	2	2	3	5	6	8	9	
	3	5	6	8	9	11	12	
	4	4	5	7	8	10	11	

$(n_{en} = 4)$

$$A = IEN(a, e)$$

LM array:

		Element numbers (e)						
		1	2	3	4	5	6	$(n_{el} = 6)$
Local node numbers (a)	1	0*	1	0	3	0	5	
	2	1	2	3	4	5	6	
	3	3	4	5	6	7	8	
	4	0	3	0	5	0	7	

$(n_{en} = 4)$

$$P = LM(a, e) = ID(IEN(a, e))$$

* Temperature boundary conditions ("q-type") denoted by zeros.

Figure 2.6.2. ID, IEN, and LM arrays for the mesh of Fig. 2.6.1.

In terms of the IEN and LM arrays, a precise definition of the q_a^e 's may be given (see (2.5.8)):

$$q_a^e = \begin{cases} 0 & \text{if } \text{LM}(a, e) \neq 0 \\ q_A & \text{if } \text{LM}(a, e) = 0, \text{ where } A = \text{IEN}(a, e) \end{cases} \quad (2.6.3)$$

This definition may be easily programmed.

In our final example of this section we shall illustrate the assembly procedure for a typical element subjected to essential boundary conditions.

Example 2

Consider a typical, four-noded element e . Assume values of the LM array, for this element, are given as follows:

$$\left. \begin{aligned} \text{LM}(1, e) &= 5 \\ \text{LM}(2, e) &= 0 \\ \text{LM}(3, e) &= 0 \\ \text{LM}(4, e) &= 9 \end{aligned} \right\} \quad (2.6.4)$$

We deduce from (2.6.4) that the contributions to the global arrays are given as follows:⁸

$$\left. \begin{aligned} K_{55} &\leftarrow K_{55} + k_{11}^e \\ K_{59} &\leftarrow K_{59} + k_{14}^e \\ K_{95} &\leftarrow K_{95} + k_{41}^e \\ K_{99} &\leftarrow K_{99} + k_{44}^e \end{aligned} \right\} \quad (2.6.5)$$

$$\left. \begin{aligned} F_5 &\leftarrow F_5 + f_1^e \\ F_9 &\leftarrow F_9 + f_4^e \end{aligned} \right\} \quad (2.6.6)$$

Note that all terms in the second and third rows and columns of k^e do *not* contribute to K . However, they may contribute to F via f_1^e and f_4^e , since, by (2.5.8) we have

$$f_1^e = \cdots - k_{12}^e q_2^e - k_{13}^e q_3^e \quad (2.6.7)$$

$$f_4^e = \cdots - k_{42}^e q_2^e - k_{43}^e q_3^e \quad (2.6.8)$$

in which, for clarity, we have omitted the first two terms of the right-hand side of (2.5.8).

Special subroutines are easily programmed to carry out the operations indicated in (2.6.5)–(2.6.8).

It is instructive to visualize the contributions of the e th element to the global stiffness and force. These contributions are depicted in Fig. 2.6.3.

We note that all necessary element assembly information is provided by the LM array.

⁸ Due to symmetry, k_{41}^e is not actually assembled in practice.

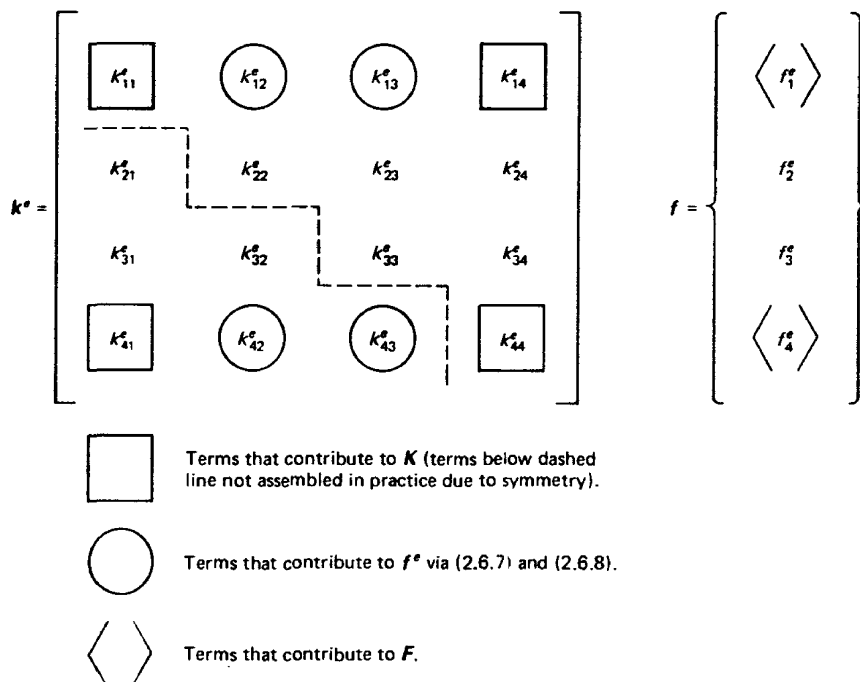
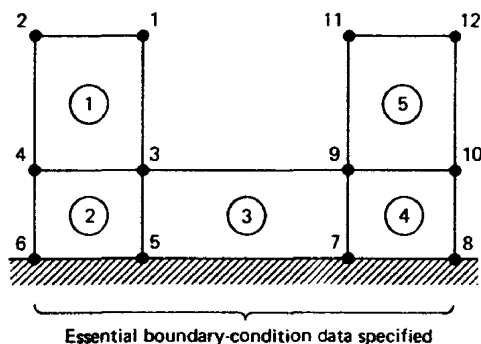


Figure 2.6.3 Contributions of heat conduction element in Example 2 to global arrays.

Exercise 1. Consider the accompanying mesh. Set up the ID, IEN, and LM arrays.



2.7 CLASSICAL LINEAR ELASTOSTATICS: STRONG AND WEAK FORMS; EQUIVALENCE

Classical elastostatics is a rich subject in its own right. See [3–7] for background and references to the literature. These references range from the very physical to the very